

Richard Prange

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EDUCATION

Western Carolina University

Bachelor of Science in Computer Science

Cullowhee, NC

May, 2025

Western Carolina University

Bachelor of Science in Mathematics

Cullowhee, NC

May, 2025

University of North Carolina at Charlotte

Masters of Science in Mathematics

Charlotte, NC

Aug, 2025 → Present

Relevant Courses: Data Structures and Algorithms, Software Development & Engineering, Problem Solving and Programming 1 & 2 (OOP), Cloud Computing, Information Security, Computer Networking, Operating Systems

EXPERIENCE

Mathematics and Computer Science Tutor

Western Carolina University

Cullowhee, NC

January, 2022 → May, 2025

- Provide personalized assistance for drop in students
- Foster critical thinking and problem solving skills through active learning techniques
- Helped students learn and retain abstract mathematics and computer science topics, leading to an increase in re-enrollment rates

PROJECTS

WheeBFit | React, TypeScript, Node.js, Express, JWT, Heroku, SQL

May, 2025

- Developed and deployed a full stack web application for a client
- Implemented a highly interactive student interface designed to keep students entertained along with a dashboard which allows teachers to track, display, and monitor a class' progress
- Information is stored in a SQL database and is accessible to authorized users
- **Available at:** <https://wheebfit.com>

ChessMeister | Java, JavaFX

May, 2024

- Developed a chess program that allows 2-4 players to play chess through a CLI or GUI
- Implemented valid check and checkmate algorithms, used to end the game and validate moves
- Incorporated functionality to allow for users to undo and redo moves
- Designed a loosely coupled system using interfaces and model view controller to allow for seamless future expansion on the user interface
- **Available at:** <https://gitfront.io/r/ricwp12/6W3trXvVkj6g/ChessMeister/>

Rusty Battleship | Rust

January, 2025

- Developed a program that allows N-players to connect to a server and play a CLI version of BattleShip
- Provided functionality using threads that allow users to continuously receive updates from the server while issuing commands via the command line to view the game state or make a move
- Implemented shared memory to allow the server to spawn threads for each user which efficiently manages user requests, messages, and state updates
- **Available at:** <https://gitfront.io/r/ricwp12/U24fPLcj6QfM/Battleship/>

GraphAnalyzer | Java

January, 2024

- Developed a program that converts .txt files to undirected graphs and allows users to perform operations on the graph.
- Implemented algorithms to perform **Depth First Search**, **Breadth First Search**, and **Transitive Closure**
- **Available at:** <https://gitfront.io/r/ricwp12/bwKiUikCwy6Q/GraphAnalyzer/>

TECHNICAL SKILLS

Languages: Python, Java, R, C++, C, JavaScript, Rust, SQL, TypeScript

Frameworks/Tools: Node.js, Express.js, TailwindCSS, SFML, React

Other: Linux, Docker, Heroku, Google Cloud Platform, Git